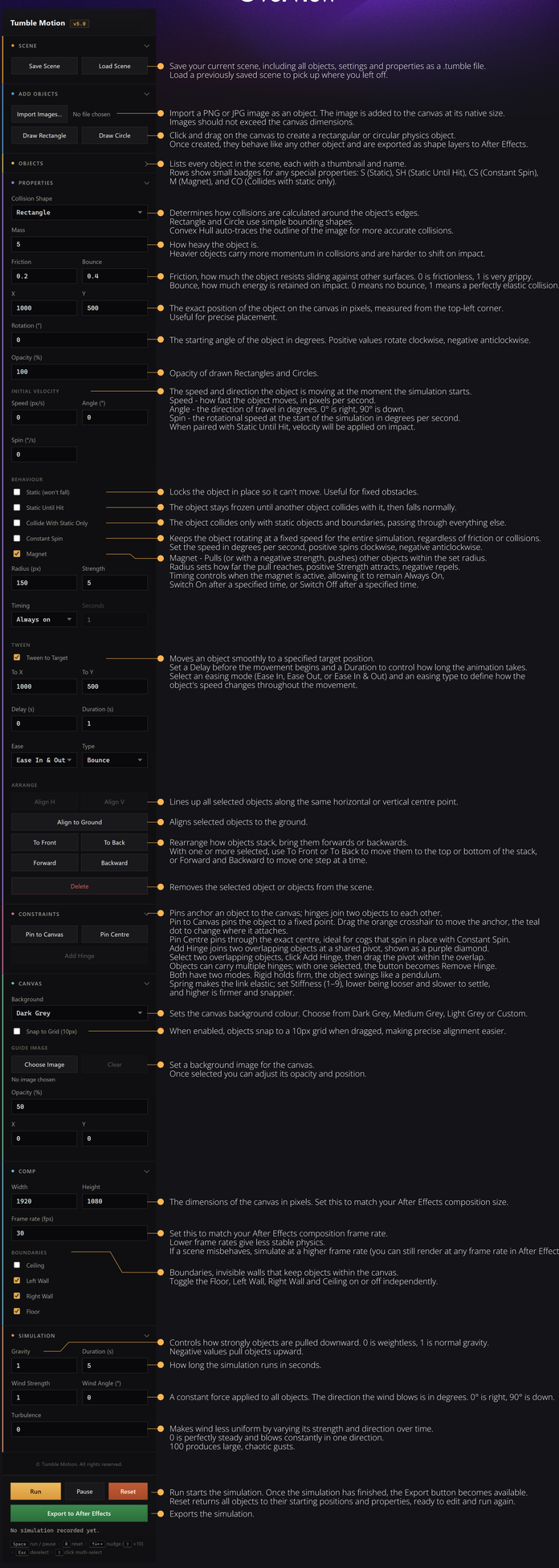
# Tumble Motion V5 Overview

Quick Guide




Save your current scene, including all objects, settings and properties as a .tumble file.
Load a previously saved scene to pick up where you left off.

Import a PNG or JPG image as an object. The image is added to the canvas at its native size.
Images should not exceed the canvas dimensions.

Click and drag on the canvas to create a rectangular or circular physics object.
Once created, they behave like any other object and are exported as shape layers to After Effects.

Lists every object in the scene, each with a thumbnail and name.
Rows show small badges for any special properties: S (Static), SH (Static Until Hit), CS (Constant Spin), M (Magnet), and CO (Collides with static only).

Determines how collisions are calculated around the object's edges.
Rectangle and Circle use simple bounding shapes.
Convex Hull auto-traces the outline of the image for more accurate collisions.

How heavy the object is.
Heavier objects carry more momentum in collisions and are harder to shift on impact.

Friction, how much the object resists sliding against other surfaces. 0 is frictionless, 1 is very grippy.
Bounce, how much energy is retained on impact. 0 means no bounce, 1 means a perfectly elastic collision.

The exact position of the object on the canvas in pixels, measured from the top-left corner.
Useful for precise placement.

The starting angle of the object in degrees. Positive values rotate clockwise, negative anticlockwise.

Opacity of drawn Rectangles and Circles.

The speed and direction the object is moving at the moment the simulation starts.
Speed - how fast the object moves, in pixels per second.
Angle - the direction of travel in degrees. 0° is right, 90° is down.
Spin - the rotational speed at the start of the simulation in degrees per second.
When paired with Static Until Hit, velocity will be applied on impact.

Locks the object in place so it can't move. Useful for fixed obstacles.

The object stays frozen until another object collides with it, then falls normally.

The object collides only with static objects and boundaries, passing through everything else.

Keeps the object rotating at a fixed speed for the entire simulation, regardless of friction or collisions.
Set the speed in degrees per second, positive spins clockwise, negative anticlockwise.

Magnet - Pulls (or with a negative strength, pushes) other objects within the set radius.
Radius sets how far the pull reaches, positive Strength attracts, negative repels.
Timing controls when the magnet is active, allowing it to remain Always On,
Switch On after a specified time, or Switch Off after a specified time.

Moves an object smoothly to a specified target position.
Set a Delay before the movement begins and a Duration to control how long the animation takes.
Select an easing mode (Ease In, Ease Out, or Ease In & Out) and an easing type to define how the object's speed changes throughout the movement.

Lines up all selected objects along the same horizontal or vertical centre point.

Aligns selected objects to the ground.

Rearrange how objects stack, bring them forwards or backwards.
With one or more selected, use To Front or To Back to move them to the top or bottom of the stack,
or Forward and Backward to move one step at a time.

Removes the selected object or objects from the scene.

Pins anchor an object to the canvas; hinges join two objects to each other.
Pin to Canvas pins the object to a fixed point. Drag the orange crosshair to move the anchor, the teal dot to change where it attaches.
Pin Centre pins through the exact centre, ideal for cogs that spin in place with Constant Spin.
Add Hinge joins two overlapping objects at a shared pivot, shown as a purple diamond.
Select two overlapping objects, click Add Hinge, then drag the pivot within the overlap.
Objects can carry multiple hinges; with one selected, the button becomes Remove Hinge.
Both have two modes. Rigid holds firm, the object swings like a pendulum.
Spring makes the link elastic; set Stiffness (1–9), lower being looser and slower to settle,
and higher is firmer and snappier.

Sets the canvas background colour. Choose from Dark Grey, Medium Grey, Light Grey or Custom.

When enabled, objects snap to a 10px grid when dragged, making precise alignment easier.

Set a background image for the canvas.
Once selected you can adjust its opacity and position.

The dimensions of the canvas in pixels. Set this to match your After Effects composition size.

Set this to match your After Effects composition frame rate.
Lower frame rates give less stable physics.
If a scene misbehaves, simulate at a higher frame rate (you can still render at any frame rate in After Effects).

Boundaries, invisible walls that keep objects within the canvas.
Toggle the Floor, Left Wall, Right Wall and Ceiling on or off independently.

Controls how strongly objects are pulled downward. 0 is weightless, 1 is normal gravity.
Negative values pull objects upward.

How long the simulation runs in seconds.

A constant force applied to all objects. The direction the wind blows is in degrees. 0° is right, 90° is down.

Makes wind less uniform by varying its strength and direction over time.
0 is perfectly steady and blows constantly in one direction.
100 produces large, chaotic gusts.

Run starts the simulation. Once the simulation has finished, the Export button becomes available.
Reset returns all objects to their starting positions and properties, ready to edit and run again.

Exports the simulation.

# Importing into After Effects

1. Run your simulation in Tumble Motion and click **Export to After Effects**.
   Save the .jsx file.
2. Open After Effects and create a new composition, making sure the dimensions and frame rate match your Tumble Motion settings.
3. Import the same assets you used in Tumble Motion and drag them into the composition.
4. Make sure the composition is open and active in the timeline.
5. Go to **File** > **Scripts** > **Run Script File** and select the exported .jsx file.
6. Your layers will have Position and Rotation keyframes applied automatically.
   You may need to reorder your layers in the timeline.

** Tumble Motion uses a real-time physics simulation. While the default settings are designed to produce stable, realistic results, extreme values (such as very high forces, masses, velocities, or gravity) may lead to unstable or unpredictable behaviour. If a simulation behaves unexpectedly, try reducing the values or running the simulation with more moderate settings.*